

UNIT-4

SEMICONDUCTOR DEVICES

Short Answer type Questions

1. What is a PN junction diode, and what is the depletion layer in a PN junction diode?
2. Define forward biasing & reverse biasing of a PN junction diode.
3. What is the effect of forward bias & reverse bias on the flow of current?
4. What is a potential barrier & how does it vary with temperature?
5. What is reverse saturation current in a PN junction diode?
6. Why does the current increase rapidly in forward bias after a certain voltage?
7. What happens if the reverse bias voltage is increased beyond the breakdown voltage?
8. Give one application each of forward bias and reverse bias of a diode
9. What is a Bipolar Junction Transistor (BJT)?
10. How many layers and junctions does a transistor have?
11. Which terminal of a BJT is always heavily doped? What are the three terminals of a BJT?
12. What is the function of the emitter, base, and collector?
13. What are the biasing conditions required for transistor action?
14. Which configuration gives **high current gain** & high voltage gain?
15. Which configuration has **high input resistance and low output resistance**?
16. Which configuration is most widely used and why?
17. Which junction is forward-biased and which is reverse-biased in normal operation?
18. Why does a small base current produce a large collector current?
19. Define current gain in CB configuration (α), and current gain in CE configuration (β).
Why is $\beta \gg \alpha$ in a transistor?
20. State the typical range of β for a transistor.
21. How is voltage gain expressed in terms of current gain and resistance?
22. Why is the voltage gain of the CC configuration less than 1?
23. What are the main terminals of a Field Effect Transistor (FET)?
24. Differentiate between n-channel and p-channel FETs in terms of construction.
25. What is the function of the gate in a JFET?
26. What happens to the channel width when reverse bias at the gate is increased?
27. Why is FET called a **voltage-controlled device**? Why does a FET exhibit unipolar conduction?
28. How does applying a negative gate voltage control the drain current in a JFET?
29. Define the “pinch-off voltage” in a JFET. Explain why the drain current saturates after pinch-off.
30. What are the two types of MOSFETs based on mode of operation?
31. Why is the input resistance of a MOSFET extremely high? How does the channel differ in depletion-type and enhancement-type MOSFETs?
32. Which MOSFET is normally ON and which is normally OFF?
33. How does a depletion-type MOSFET work in enhancement mode?

34. How does an enhancement-type MOSFET begin to conduct.

Long Answer Type Question

1. Define n type and p-type semiconductors.. What is a p-n junction diode? Discuss its operation. Also, describe the biasing of a p-n junction diode.
2. What are bipolar junction transistors? Explain the mechanism of current flow in PNP and NPN transistors. Discuss the working of a BJT as an amplifier in a common-base configuration. Define the current gain in the common-base configuration. Derive the expressions for voltage gain and power gain.
3. Discuss the working of a BJT as an amplifier in common-emitter configuration. Define the current gain in the common-emitter configuration. Derive the expressions for voltage gain and power gain.
4. Explain the construction and working principle of a field-effect transistor. Describe the basic structure, including the n-channel or p-channel, source, drain, gate terminals, and how the depletion region is formed.
5. Describe the construction of a MOSFET, detailing the significance of the metal, oxide, and semiconductor layers in its operation.
6. Discuss the difference between Enhancement-mode and Depletion-mode MOSFETs, explain the conditions for their creation and operation, and the key structural differences that allow them to function differently.
7. Explain the fundamental principle of the MOSFET. Describe how an electric field generated by the gate voltage controls the flow of current between the source and the drain terminals, emphasizing why it is a 'voltage-controlled' and 'unipolar'.

Numerical Problems

1. The saturation current density of a p-n junction diode of germanium crystal is 300 mA/m^2 at 300 K. Find the voltage that would have to be applied to cause a forward current density of 10^5 A/m^2 to flow. Ans- 0.329 V
2. The saturation current density of a p-n junction diode of germanium crystal is 200 mA/m^2 , at 300 K. Find the voltage that would have to be applied to cause a forward current density of $5 \times 10^4 \text{ A/m}^2$ to flow.
Ans - 0.322 V
3. The saturation current density of a p-n junction diode of silicon crystal is 300 mA/m^2 , at 300 K. Find the voltage that would have to be applied to cause a forward current density of 10^5 A/m^2 to flow.
Ans- 0.658 V

4. In a BJT, when the base current is increased by $50\text{ }\mu\text{A}$, keeping the collector voltage fixed at 2 volts, the collector current increases by 1 mA. Calculate the current gain α and β .

Ans – 20, 0.95

5. Current gain of a BJT in common-base configuration is 0.88. Find the value of base - current when the emitter - current is 2 mA.

Ans- 0.24 mA

6. The reverse saturation current in a BJT in common base configuration is $12.5\text{ }\mu\text{A}$. For an emitter current of 2 mA, the collector current is 1.97 mA. Find the current gain and the base current.

Ans - 0.03 mA

7. In a BJT, $\alpha = 0.98$ and $I_{CBO} = 5\text{ }\mu\text{A}$. Determine current gain β . Also, find the value of leakage current I_{CEO} .

Ans - 0.25 mA

8. A PNP transistor in common-base configuration offers a dynamic input resistance of $50\text{ }\Omega$ and a current gain of 0.96. If the load resistance in the collector circuit is $5\text{ k}\Omega$, find the voltage gain and power gain.

Ans – 92

9. Find the value of β if i) $\alpha = 0.9$ ii) $\alpha = .94$

Ans – 9, 15.67

10. Calculate the value of emitter current (I_E) for which $\beta = 40$ and $I_B = 25\text{ }\mu\text{A}$.

Ans – 1.025

11. For an n-channel JFET

$I_{DSS} = 20\text{ mA}$, $V_P = -8\text{ V}$, $g_{mo} = 5000\text{ }\mu\text{S}$ Find the value of drain current I_D and trans conductance at $V_{GS} = -4\text{ V}$.

Ans – 5 ma, $2500\text{ }\mu\text{S}$

12. Determine the value of transconductance of a FET when the drain current changes from 1 mA to 1.9 mA with a change in gate-source voltage from - 3.3V to -3V.

Ans- - 0.3 volts, $3\text{ }\frac{\text{mA}}{\text{V}}$

13. For a FET $I_{DSS} = 8.4\text{ mA}$, $V_P = -3\text{ V}$, what is the value of I_D for $V_{GS} = -1.5\text{ V}$? Find g_m at this point.

Ans - $2.8\text{ }\frac{\text{mA}}{\text{V}}$

14. An n-channel JFET has a pinch-off voltage of -4.5V and $I_{DSS} = 9\text{ mA}$. At what value of V_{GS} will I_D equal 3mA? What is its g_m at this I_D ?

Ans - 1.902 volts, $2.31 \frac{mA}{V}$

15. A JFET has $V_P = -4.5V$, $I_{DSS} = 10mA$ and $I_D = 2.5mA$. Determine the transconductance (g_m).

Ans - $2.22 \frac{mA}{V}$

16. For an n-channel JFET $I_{DSS} = 8.7mA$, $V_P = -3V$ $V_{GS} = -1V$.

Find the value of i) I_D ii) g_m

Ans - 5.8 MS, $3.86 \frac{mA}{V}$